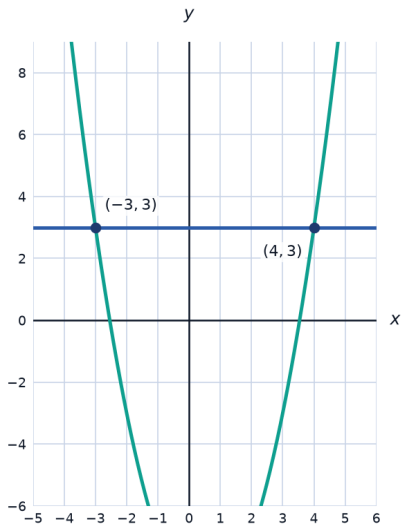
Desmos Graphing Tactics, Regression & Systems Strategy

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

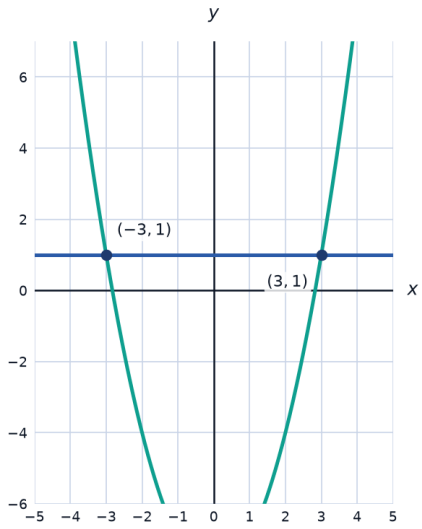
The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the community theater are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?



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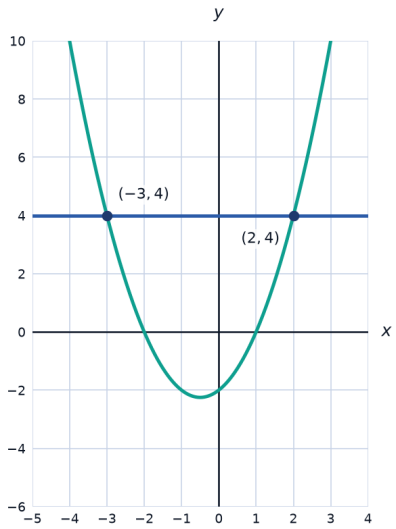
2 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the mountain observatory are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?



3 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

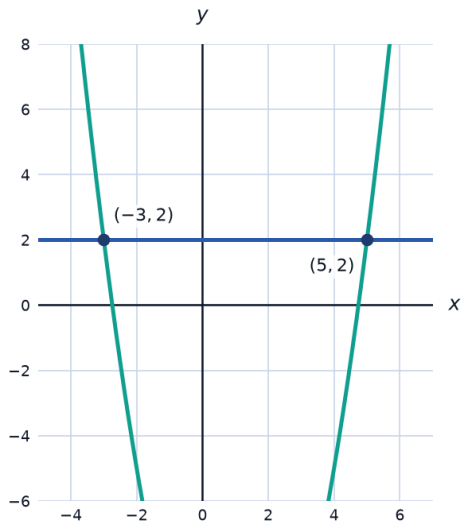
The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the solar microgrid facility are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?



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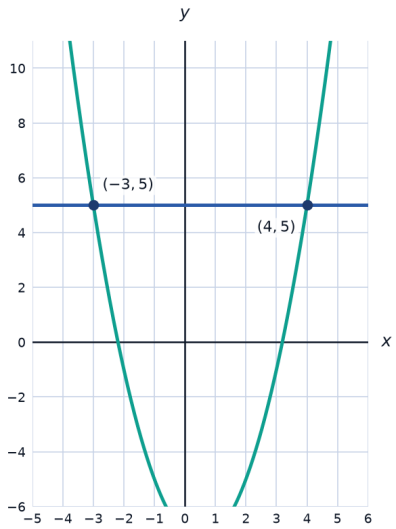
4 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the publishing house are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?



5 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

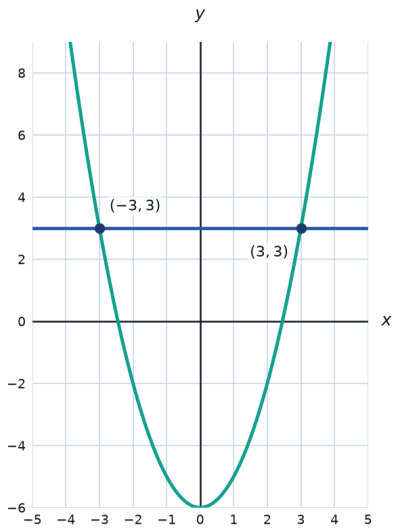
The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the hydroponic greenhouse are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?



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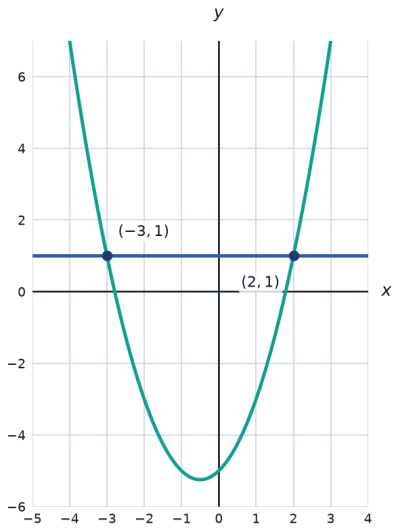
6 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the robotics club are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?



7 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the wildlife sanctuary are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?



Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

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8 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the robotics club, consider the quadratic function $p(t) = 12t^2 - 13$ over the closed interval $[13, 28]$. What is the average rate of change $\Delta p/\Delta t$ on $[13, 28]$?

9 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the meteorological station, let the functions h and k be defined by $h(u) = 22u - 7$ and $k(u) = u^2 + 3$. What is the composite value $h(k(10))$?

10 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the urban bike-share station, suppose in a hypothetical 50-question section, a student answers 90% of the first 30 questions correctly, then 60% of the remaining 20 correctly. How many total questions did the student answer correctly?

11 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the sports complex, consider the quadratic function $h(u) = 8u^2 - 15$ over the closed interval $[3, 22]$. What is the average rate of change $\Delta h/\Delta u$ on $[3, 22]$?

12 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the wind energy cooperative, since $4^3 = 64$ and $5^3 = 125$, a problem asks which integer choice is closest to $\sqrt[3]{100}$. Choices: A) 5, B) 4, C) 10, D) 20.

13 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the astronomical society, in the xy-plane, the graph of $y = (x - 2)^2 - 16$ at the municipal library intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

14 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the film archive, in the xy-plane, a circle with equation $(x - 7)^2 + (y - 9)^2 = 64$ represents a boundary at the mountain observatory. What is the greatest value of b for which the horizontal line $y = b$ is tangent to the circle?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the regional airport shuttle, in the xy-plane for the water treatment plant model, the parabola $y = x^2 + 6x + k$ and the horizontal line $y = 6$ intersect at exactly one point. What is the value of the constant k?

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16 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane for the botanical conservatory model, the parabola $y = x^2 + 4x + k$ and the horizontal line $y = 3$ intersect at exactly one point. What is the value of the constant k?

17 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the graph of $y = (x - 2)^2 - 16$ at the municipal library intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

18 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, a circle with equation $(x - 4)^2 + (y - 6)^2 = 25$ represents a boundary at the community theater. What is the greatest value of b for which the horizontal line $y = b$ is tangent to the circle?

19 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane for the regional bakery model, the parabola $y = x^2 + 10x + k$ and the horizontal line $y = 6$ intersect at exactly one point. What is the value of the constant k?

20 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

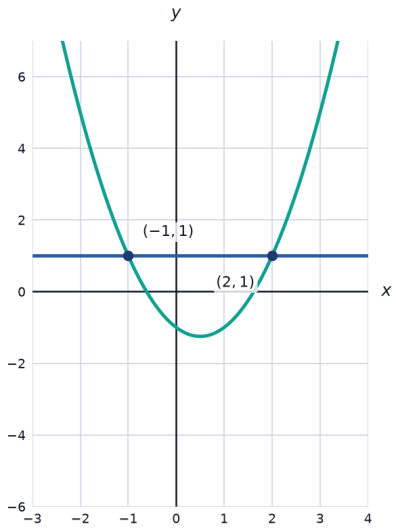
In the xy-plane, the graph of $y = (x - 5)^2 - 49$ at the coastal aquarium intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the botanical conservatory are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?



a. 3

b. 5

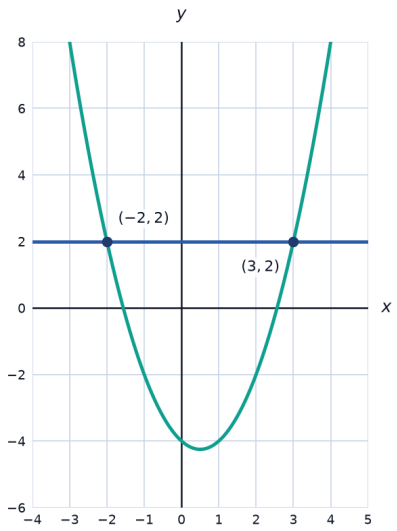
c. 2

d. 6

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22 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the municipal library are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?



a. 5

b. 6

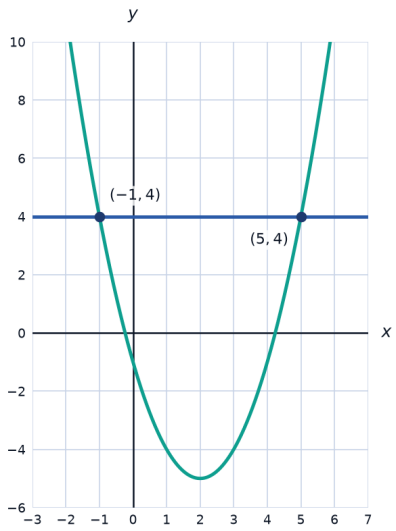
c. 2

d. 3

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23 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the regional bakery are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?



a. 7

b. 8

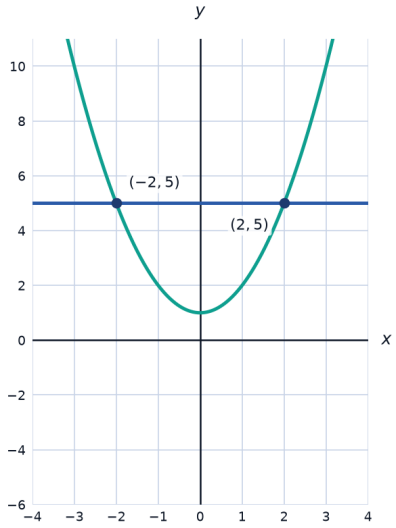
c. 10

d. 5

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24 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the coastal aquarium are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?



a. 4

b. 6

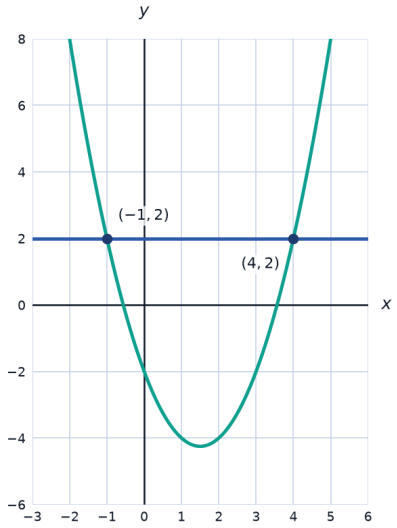
c. 3

d. 7

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25 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the science museum are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?



a. 4

b. 8

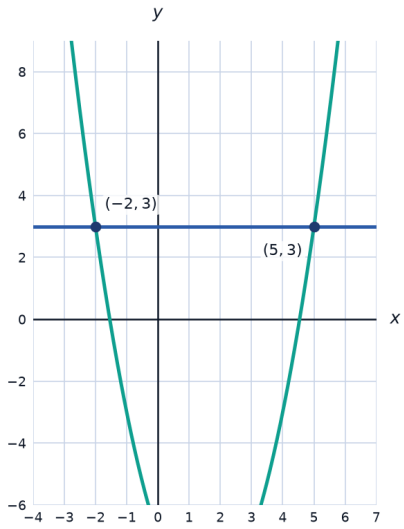
c. 5

d. 7

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26 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the urban bike-share station are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?



a. 5

b. 6

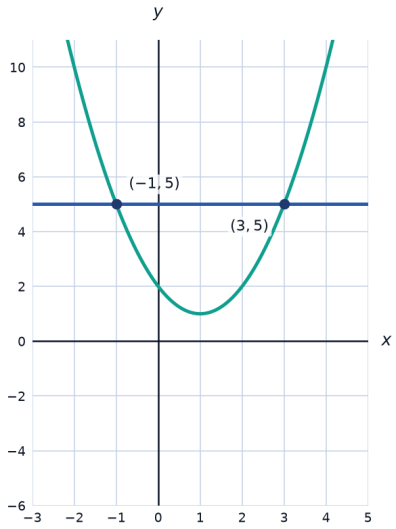
c. 8

d. 10

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27 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the ceramic studio are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?



a. 9

b. 3

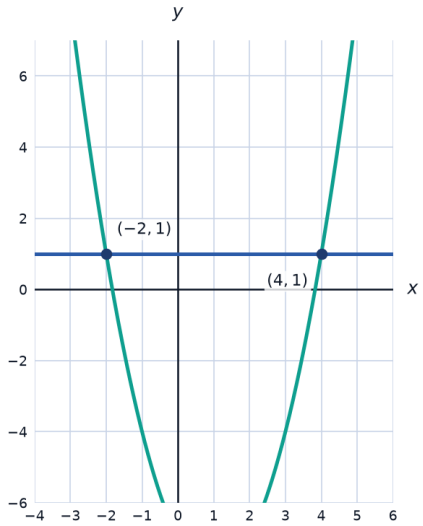
c. 6

d. 7

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28 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the agricultural cooperative are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?



a. 5

b. 9

c. 6

d. 8

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the art gallery, let the functions p and q be defined by $p(t) = 22t - 9$ and $q(t) = t^2 + 11$. What is the composite value $p(q(10))$?

a. 2444

b. 2424

c. 2455

d. 2433

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30 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the culinary institute, in the coordinate plane, the graph of $y = h(u)$ is translated 5 units right and 6 units up to form $y = k(u)$, mapping $X(2, 50)$ to Y. What is the sum of coordinates of Y on $y = k(u)$?

a. 63

b. 74

c. 58

d. 69

31 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the water treatment plant, let the functions u and v be defined by $u(s) = 16s - 12$ and $v(s) = s^2 + 6$. What is the composite value $u(v(14))$?

a. 3236

b. 3220

c. 3226

d. 3208

32 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the sports complex, in the coordinate plane, the graph of $y = p(t)$ is translated 2 units right and 15 units up to form $y = q(t)$, mapping $P(20, 67)$ to Q. What is the sum of coordinates of Q on $y = q(t)$?

a. 104

b. 119

c. 117

d. 102

33 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the maple syrup farm, suppose a student plans to spend exactly 1.5 minutes on each of 30 questions in a 50-minute section. How many total minutes would that use?

a. 50 minutes exactly

b. 45 minutes, leaving a 5-minute buffer

c. 40 minutes

d. 42 minutes

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34 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the planetarium, suppose a student answers the first 25 of 30 questions at about 1.5 minutes each, in a 50-minute section. How many minutes are left for the remaining 5 questions?

a. 15 minutes

b. 10 minutes

c. 12.5 minutes

d. 7.5 minutes

35 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the regional airport shuttle, a function f satisfies the identity $f(x-19)=22x+7$ for all real x. What is the value $f(16)$?

a. 799

b. 758

c. 784

d. 777

36 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the wind energy cooperative, suppose a student is 5 questions into a 55-minute, 27-question section after using 12 minutes, with a target pace of about 2 minutes per question. Are they on pace, ahead, or behind?

a. Behind pace (5 questions should take about 10 minutes, but 12 were used)

b. Ahead of pace

c. Exactly on pace

d. Cannot be determined

37 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the marine research vessel, suppose a student has 20 minutes left and 8 questions remaining. What's the average time available per remaining question?

a. 2 minutes

b. 2.5 minutes

c. 3 minutes

d. 1.5 minutes

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the maple syrup farm, in the xy-plane, the graph of $y = (x - 5)^2 - 16$ at the culinary institute intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

a. 13

b. 8

c. 4

d. 16

39 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the planetarium, in the xy-plane, a circle with equation $(x - 4)^2 + (y - 10)^2 = 25$ represents a boundary at the wildlife sanctuary. What is the greatest value of b for which the horizontal line $y = b$ is tangent to the circle?

a. 15

b. 5

c. 35

d. 9

40 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the wind energy cooperative, in the xy-plane, the graph of $y = (x - 3)^2 - 49$ at the orchestra hall intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

a. 49

b. 17

c. 14

d. 7

41 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the marine research vessel, in the xy-plane, a circle with equation $(x - 7)^2 + (y - 6)^2 = 64$ represents a boundary at the sports complex. What is the greatest value of b for which the horizontal line $y = b$ is tangent to the circle?

a. 15

b. 14

c. -2

d. 70

42 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the graph of $y = (x - 3)^2 - 49$ at the orchestra hall intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

a. 49

b. 17

c. 14

d. 7

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43 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, a circle with equation $(x-7)^2+(y-6)^2=64$ represents a boundary at the sports complex. What is the greatest value of b for which the horizontal line $y=b$ is tangent to the circle?

a. 15

b. 14

c. -2

d. 70

44 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane for the charter ferry service model, the parabola $y=x^2+12x+k$ and the horizontal line $y=3$ intersect at exactly one point. What is the value of the constant k?

a. 39

b. 36

c. 41

d. 42

45 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the graph of $y=(x-1)^2-100$ at the maple syrup farm intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2-x_1|$?

a. 10

b. 100

c. 21

d. 20

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